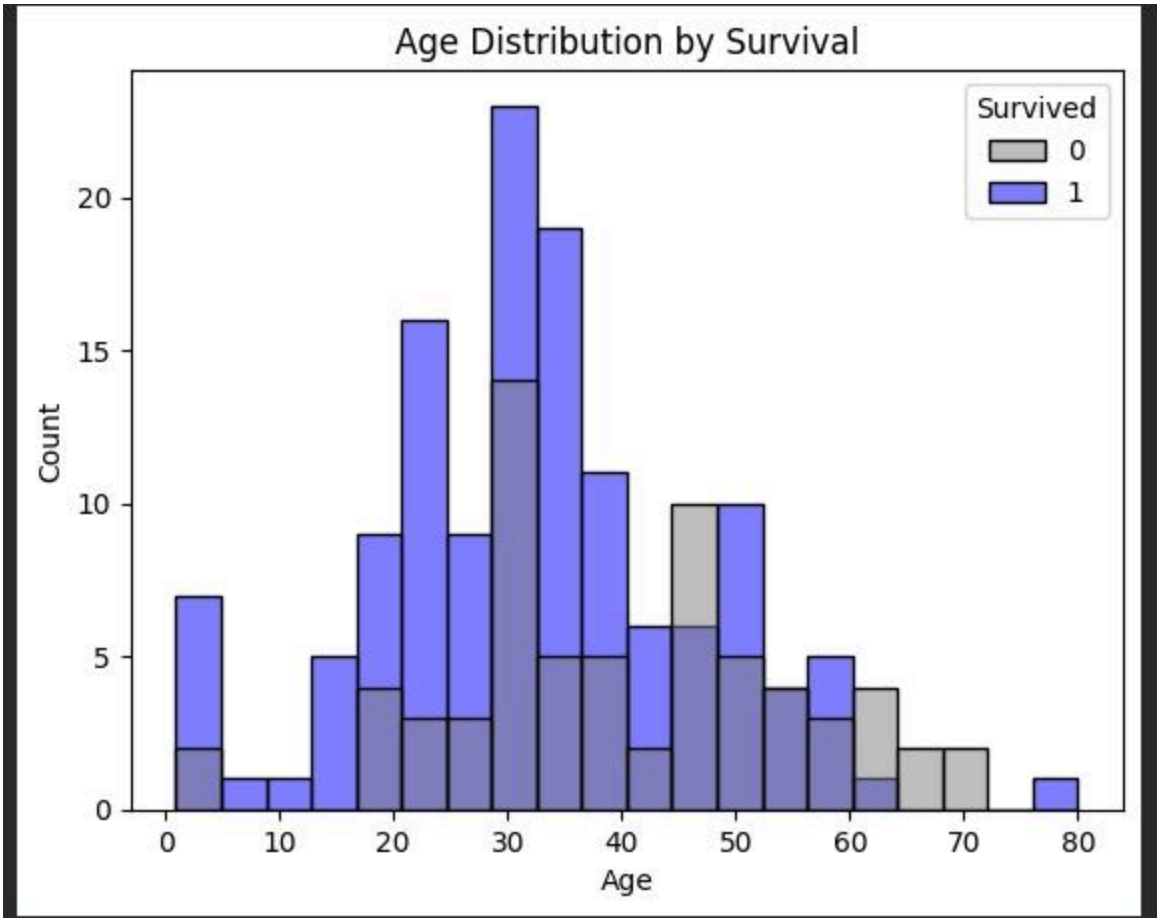
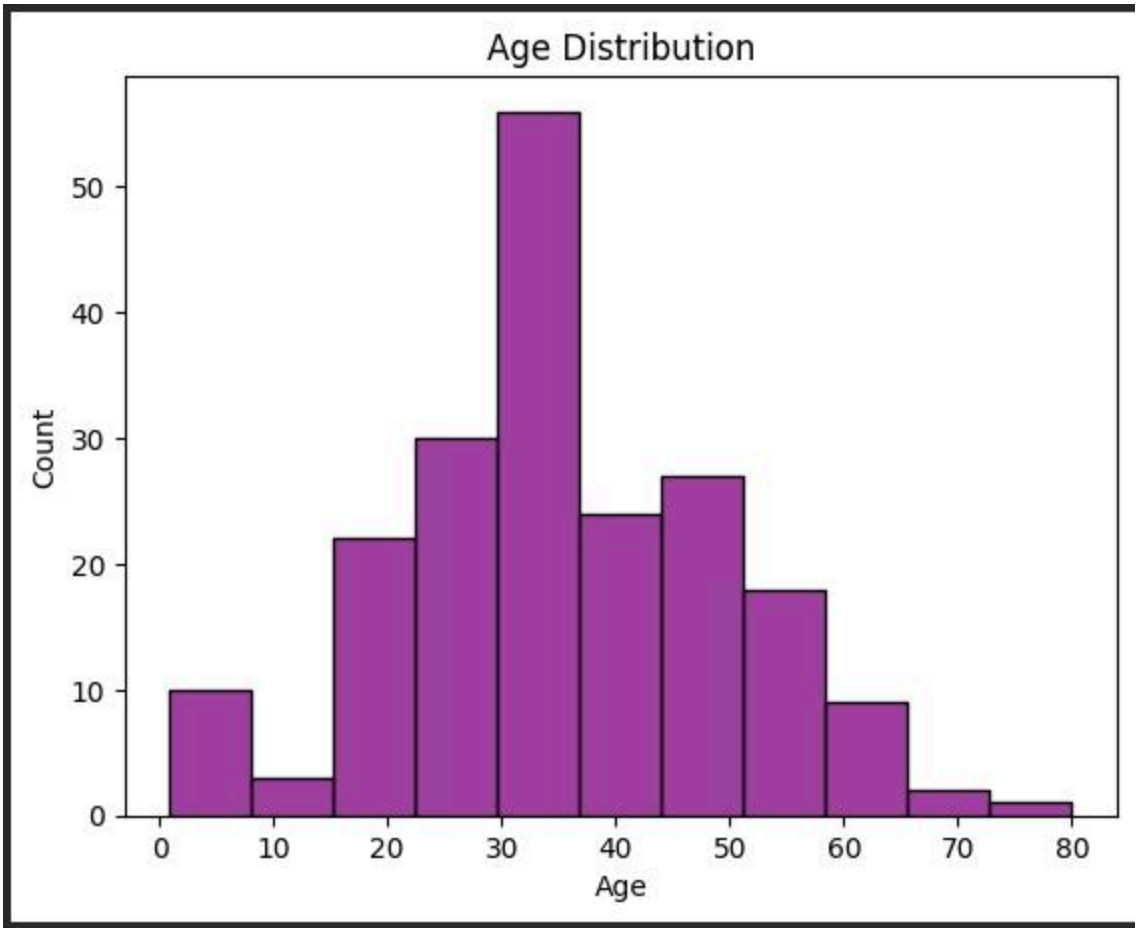
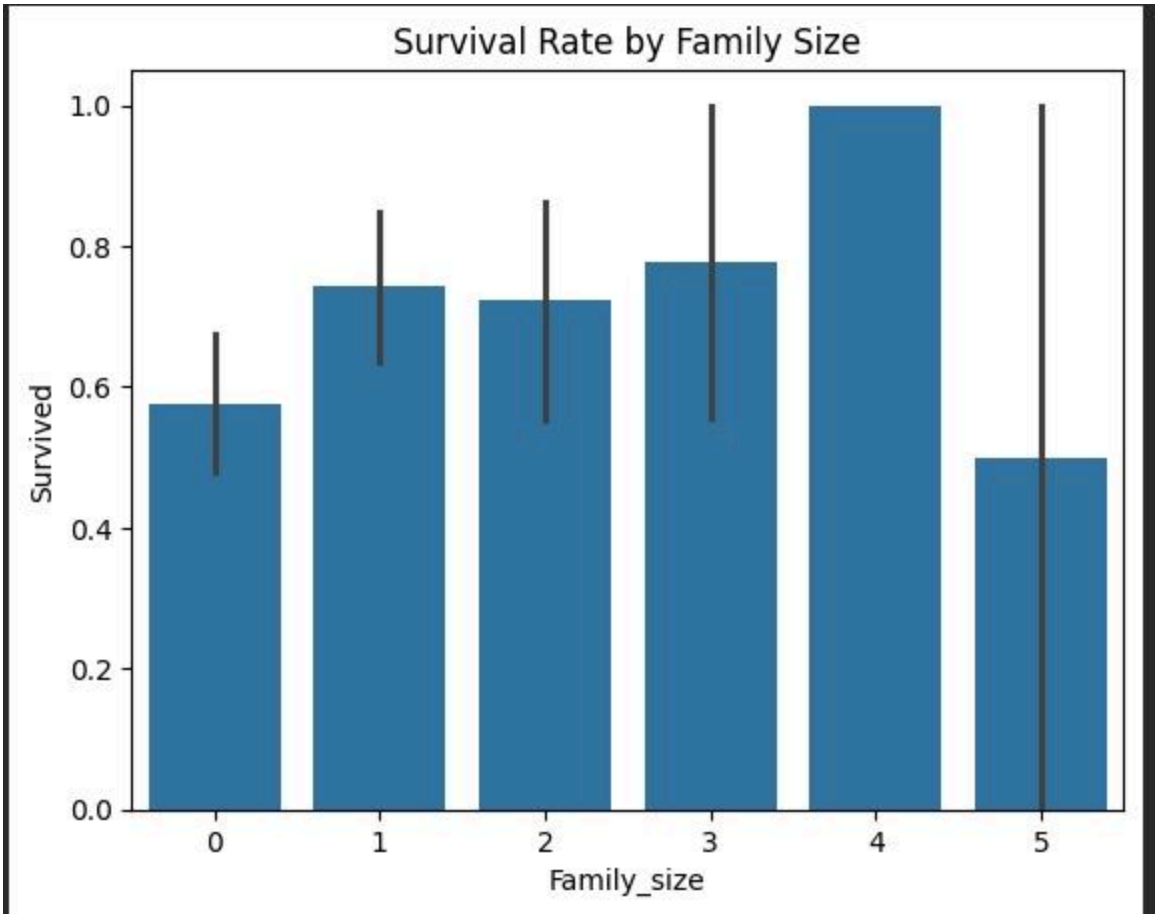
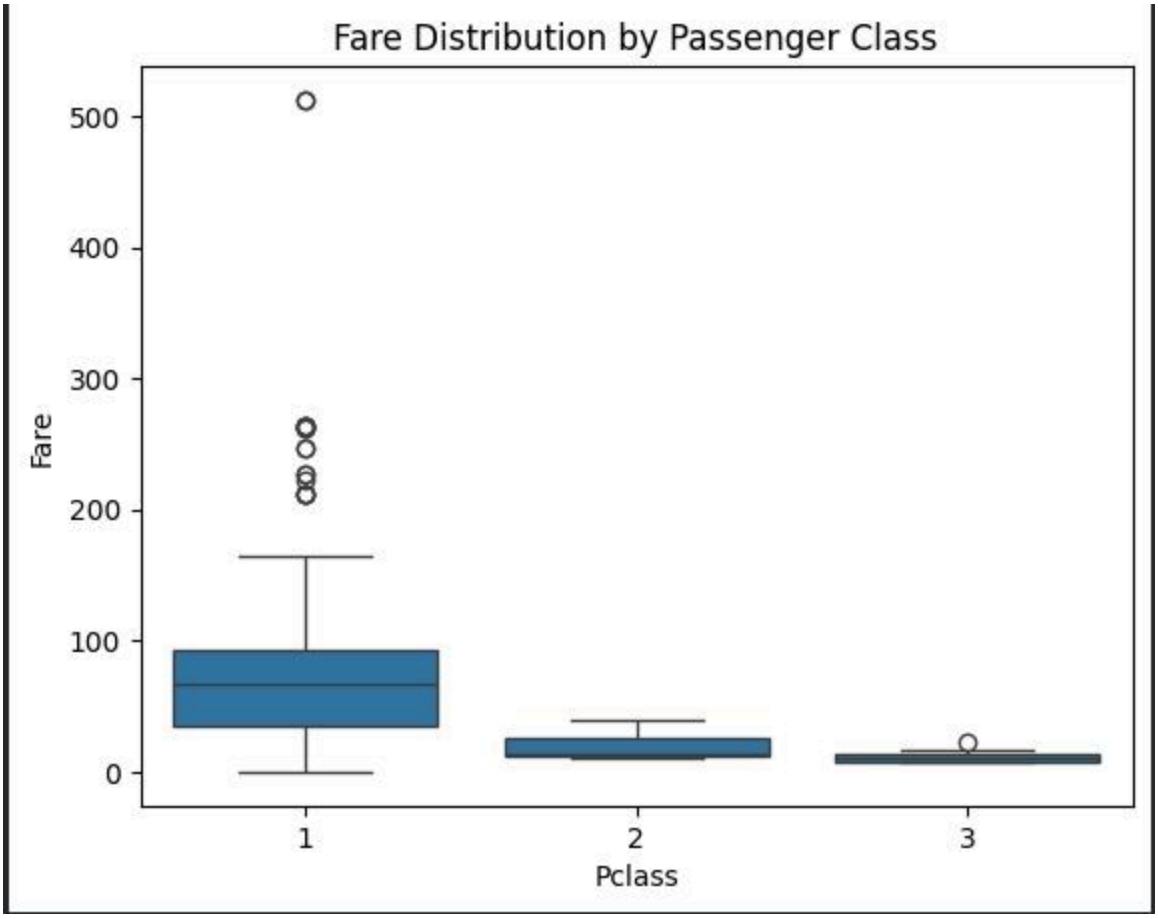
This project involves a comprehensive analysis of the Titanic dataset, covering several key stages:

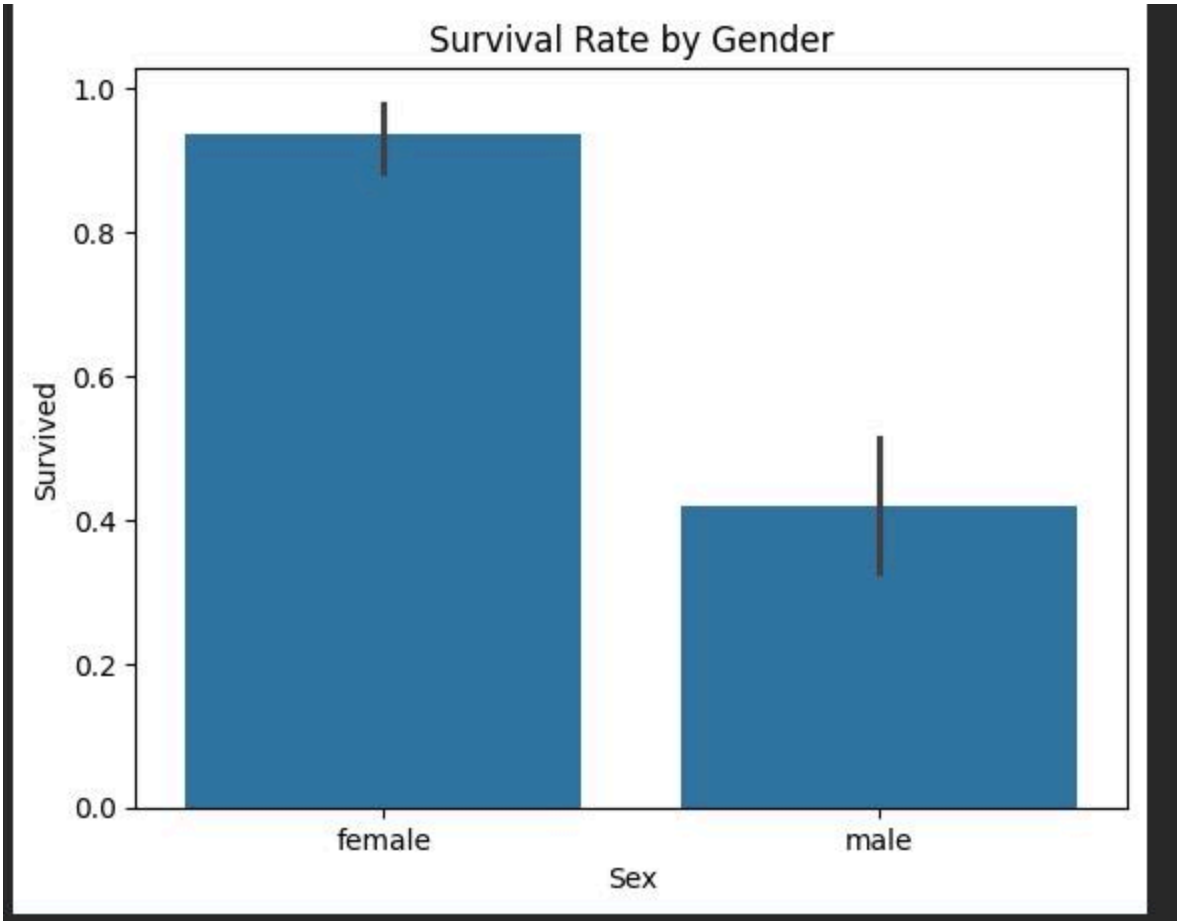
- **Data Loading and Initial Exploration:** The project starts by loading the titanic.csv file into a pandas DataFrame. Initial exploration includes viewing the first and last few rows, checking the DataFrame's shape and column names, and identifying the number of missing values in each column.
- **Data Cleaning and Preprocessing:**
  - Missing values in the 'Age' column are filled with the mean age.
  - Rows with any remaining missing values are dropped, primarily affecting the 'Cabin' and 'Embarked' columns initially.
  - Later, for modeling, missing 'Embarked' values are filled with 'S', and 'Cabin' values are filled with 'unknown' (before it's eventually dropped).
  - Categorical features like 'Sex' and 'Embarked' are converted into numerical representations using mapping.
- **Feature Engineering:** A new feature, family_size, is created by summing the 'SibSp' (siblings/spouses aboard) and 'Parch' (parents/children aboard) columns.
- **Exploratory Data Analysis (EDA) and Visualization:** Various visualizations are generated to understand the data and its relationships:
  - Overall survival rate is displayed using a bar chart and a pie chart.
  - Survival rate is analyzed by 'Sex' and 'Family_size' using bar plots.
  - The distribution of 'Fare' by 'Pclass' is shown with a box plot.
  - 'Age' distribution is visualized using histograms, both overall and differentiated by survival status.
  - A scatter plot shows the relationship between 'Age' and 'Fare'.


Age Distribution by Survival
Survived
0
1
Count
0
5
10
15
20
0
10
20
30
40
50
60
70
80
Age


Age Distribution
Count
Age
0
10
20
30
40
50
60
70
80


Survival Rate by Family Size
Survived
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Family_size


Fare Distribution by Passenger Class
500
400
300
200
100
0
Fare
1
2
3
Pclass

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# Observation: A higher percentage of females survived compared to males.
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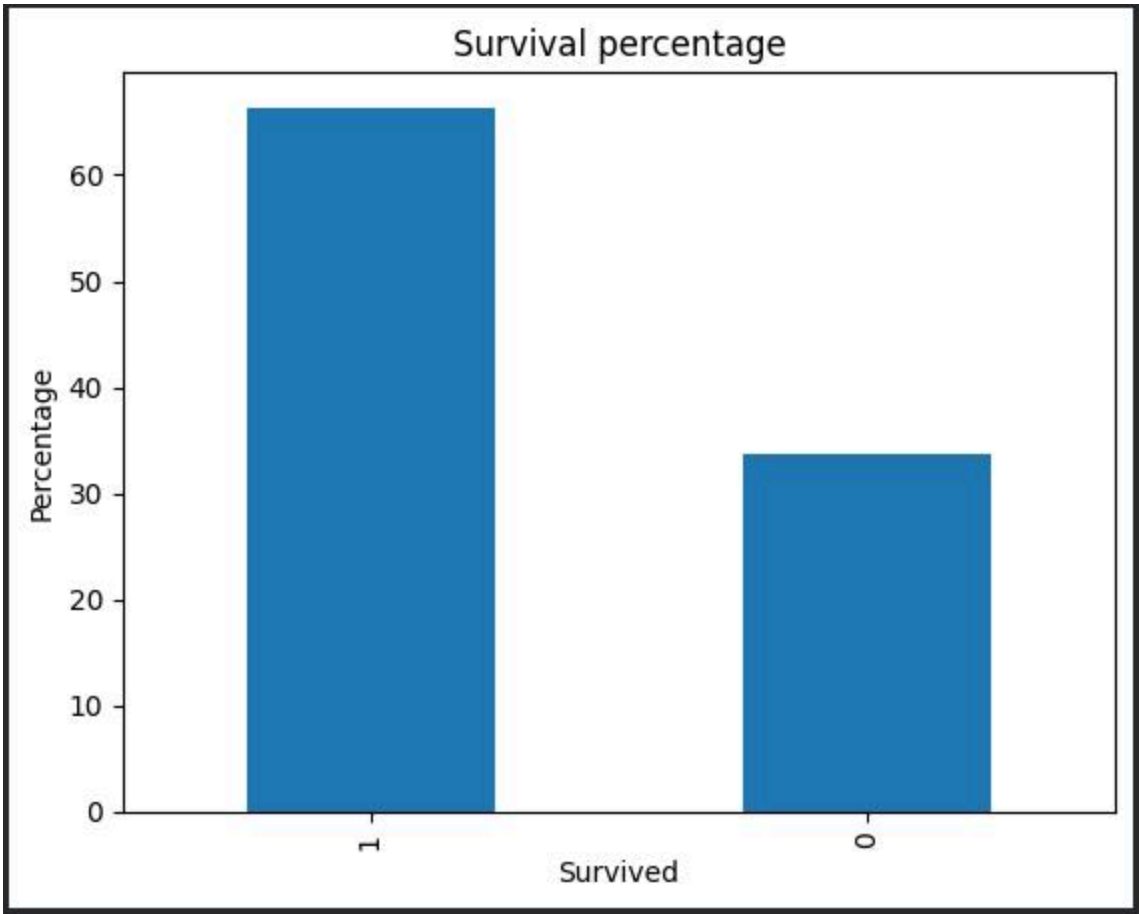
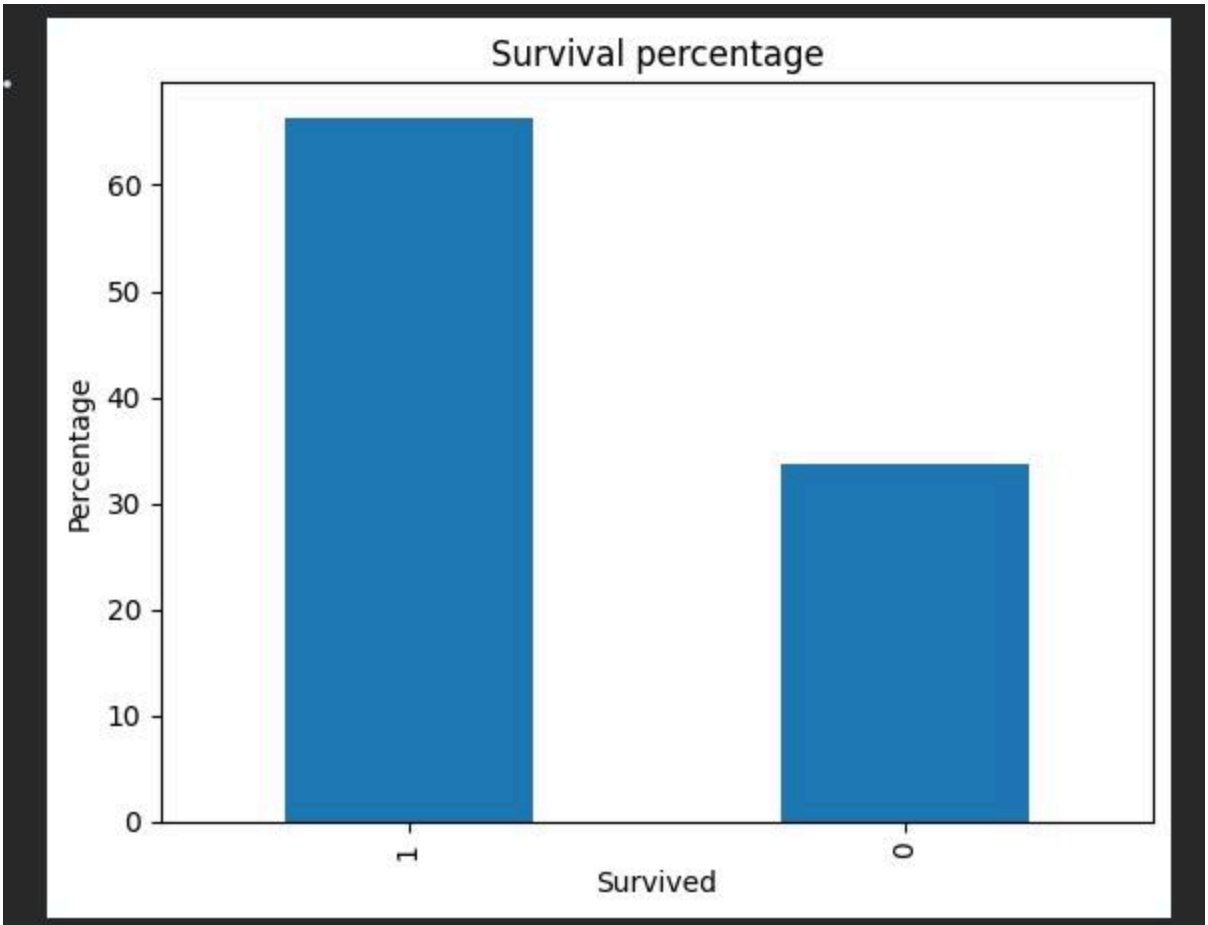


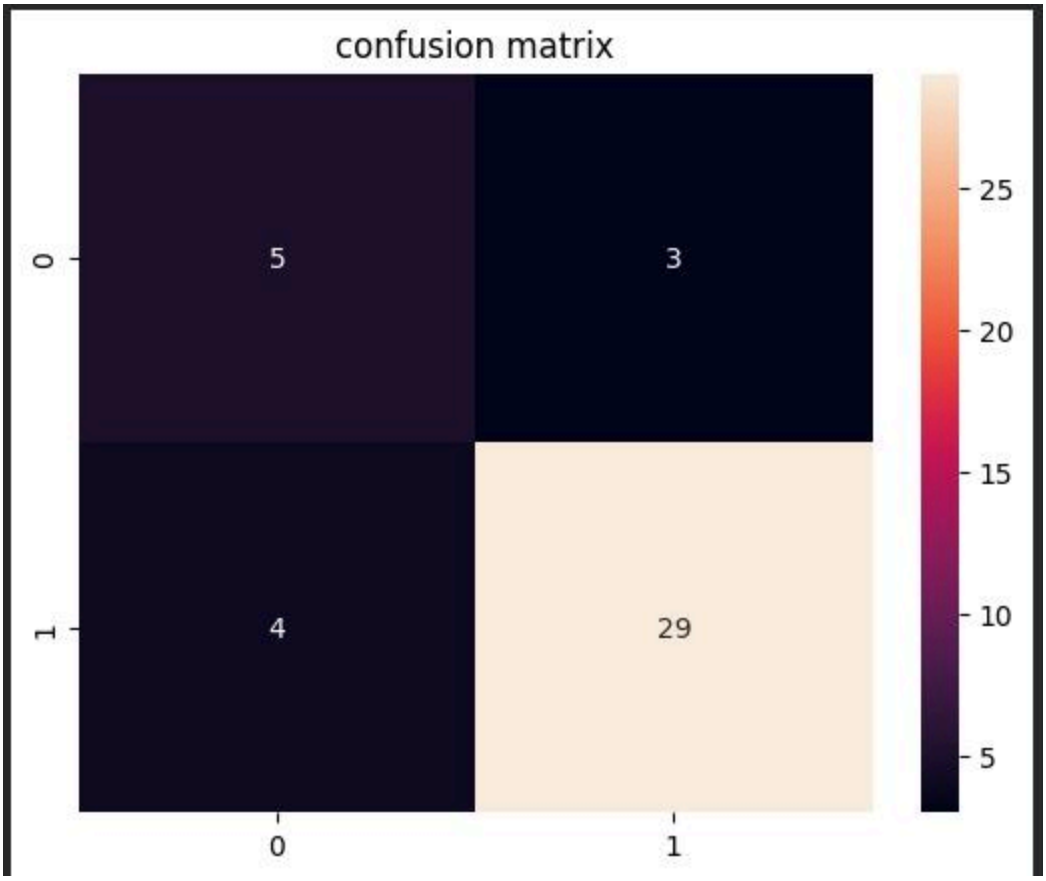

```
# The survival rate is approximately 66.3%.
```




```
# The survival rate is approximately 66.3%.
```

Machine Learning Model Training and Evaluation:

- A Logistic Regression model is used for predicting survival.
- The dataset is split into training and testing sets.
- Features (x) are defined by dropping irrelevant columns ('Survived', 'Name', 'Ticket', 'PassengerId', 'Cabin'), and 'Survived' is set as the target variable (y).
- The model is trained on the training data and then used to make predictions on the test set.
- The model's performance is evaluated using accuracy_score and a confusion_matrix.



RESULT: The model correctly identified 5 people who did not survive and 29 people who did survive. It made 3 errors by predicting survival when there was none, and 4 errors by predicting no survival when there was. This level of detail helps understand not just the overall accuracy, but also the types of errors the model is making.